

Introduction to Passband digital transmission

Digital Modulation Techniques

Passband data Transmission

→ Digital CW modulation

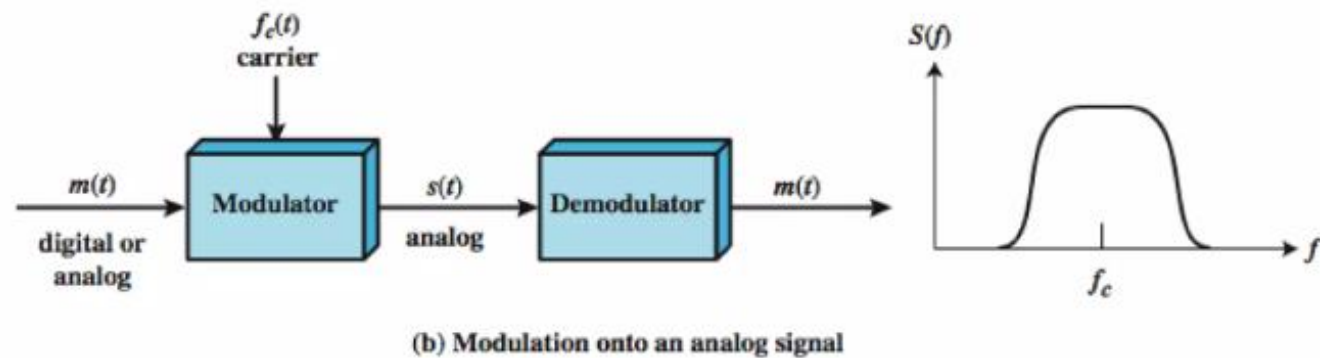
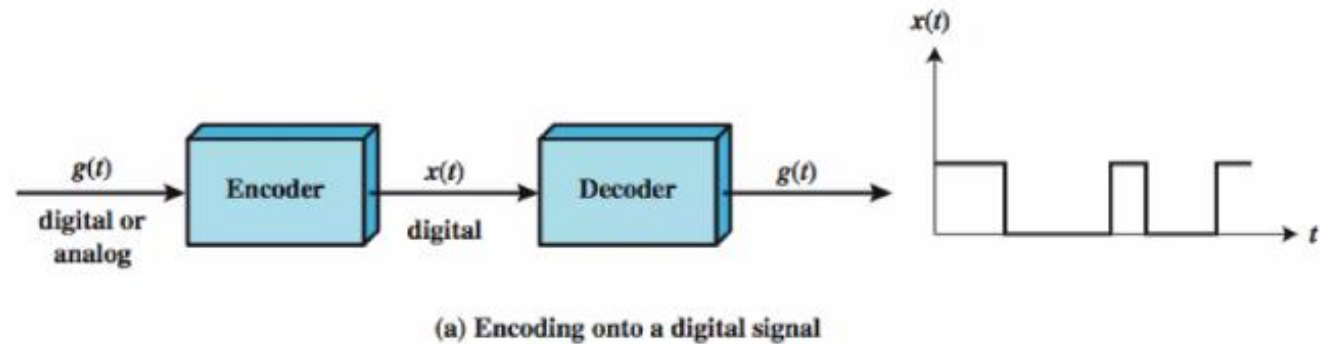
- Digital data modulates High frequency sinusoidal carrier.
- Suitable for transmission over long distance.
- Communication channel may be microwave radio link, a satellite channel or the like.

Baseband data Transmission.

- Transmission of digital data directly over the channel.
- No need of carrier or any modulation.
- Suitable for transmission over short distance.
- Communication channel using twin-wire line or Co-axial cables.

Introduction

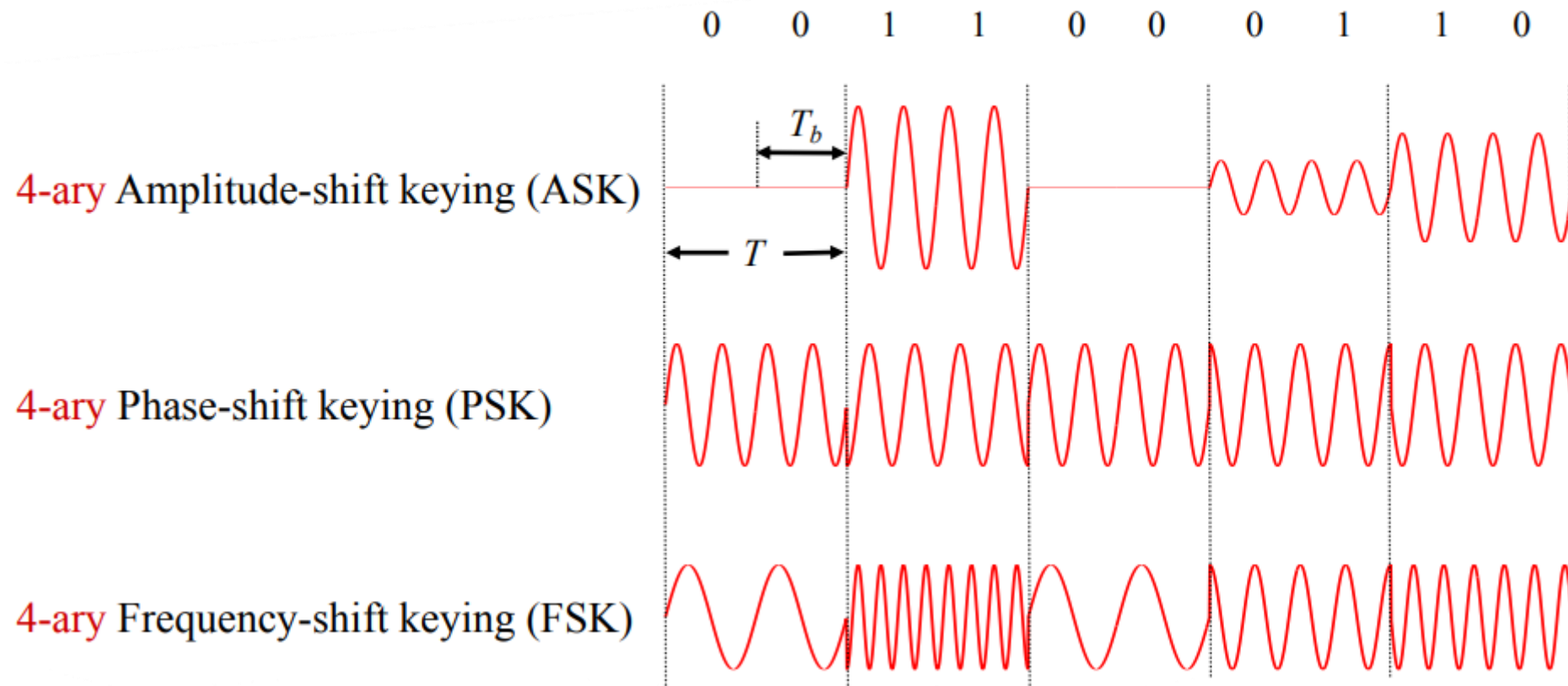
- Passband Data Transmission deals with the Transmission of the Digital Data over the real valued Passband channel



Categories of Digital Communications (ASK/PSK/FSK)

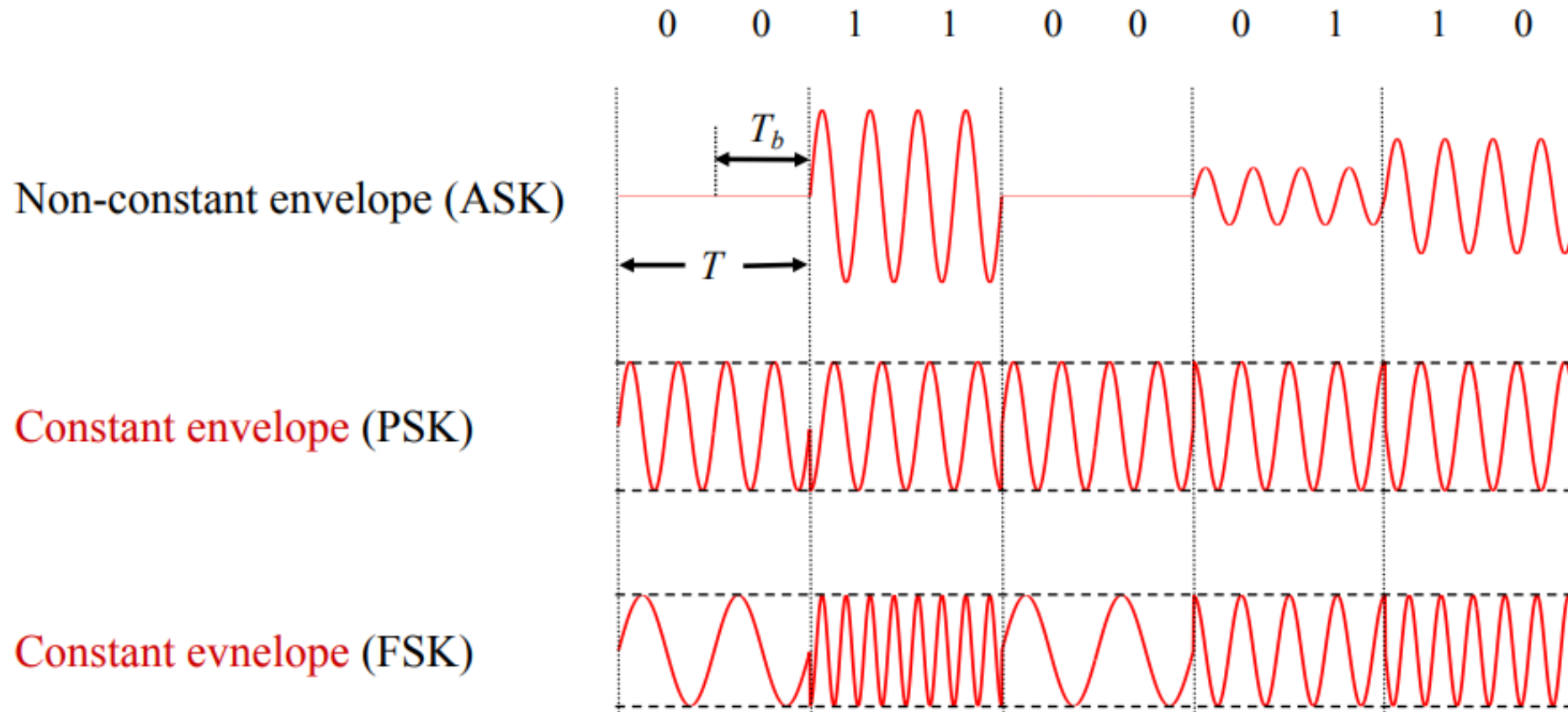
- Three basic signaling schemes in M-ary digital communications
- **Amplitude Shift Keying (ASK):** The most basic (binary) form of ASK involves the process of switching the carrier either on or off, in correspondence to a sequence of digital pulses that constitute the information signal. One binary digit is represented by the presence of a carrier, the other binary digit is represented by the absence of a carrier. Frequency remains fixed.
- **Frequency Shift Keying (FSK) :** The most basic (binary) form of FSK involves the process of varying the frequency of a carrier wave by choosing one of two frequencies (binary FSK) in correspondence to a sequence of digital pulses that constitute the information signal. Two binary digits are represented by two frequencies around the carrier frequency. Amplitude remains fixed
- **Phase Shift Keying (PSK):** Another form of digital modulation technique which involves the process of varying the phase of a carrier wave.

Categories of Digital Communications (ASK/PSK/FSK)



Categories of M-ary Digital Communications (Constant Envelope versus Non-Constant Envelope)

- Constant envelope: A necessity for non-linear channels



Categories of Digital Communications (Coherent versus Non-Coherent)

- **Coherent technique**

- The transmitter and receiver are required to be synchronized in both carrier phase and bit timing.

- **Non-Coherent technique**

- The transmitter and receiver are not required to be synchronized in both carrier phase and bit timing.

Power of Sinusoidal Signal

power of Sin/cos signals:

$$x(t) = A \sin \omega t$$

$$P = U_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 dt$$

$$= U_{T \rightarrow \infty} \frac{A^2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \left(\frac{1}{2} - \frac{1}{2} \cos 2\omega t \right) dt$$

$$= U_{T \rightarrow \infty} \frac{A^2}{2T} \left[t \right]_{-\frac{T}{2}}^{\frac{T}{2}} - U_{T \rightarrow \infty} \frac{A^2}{2T} \frac{\sin 2\omega t}{2\omega} \Bigg|_{-\frac{T}{2}}^{\frac{T}{2}}$$

$$= U_{T \rightarrow \infty} \frac{A^2}{2T} \left[\frac{T}{2} + \frac{T}{2} \right] - U_{T \rightarrow \infty} \frac{A^2}{4\omega T} \left(\sin 2\omega \frac{T}{2} + \sin 2\omega \frac{T}{2} \right)$$

$$\boxed{P = \frac{A^2}{2}}$$

$\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$
 $\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$
 $\sin n\pi = 0$
 $\omega = \frac{2\pi}{T}$
 $\boxed{\omega \cdot T = 2\pi}$

Sinusoidal Carrier

NOTE: Sinusoidal Carrier

$$c(t) = A_c \cos(2\pi f_c t + \phi_c) \text{ ----- } \underline{\text{Analog Communication}}$$

sinusoidal Power

$$P_s = \frac{1}{2} A_c^2$$

$$\text{-----} \rightarrow A_c = \sqrt{2 P_s}$$

For Digital Communication.

$$c(t) = \sqrt{2 P_s} \cos(2\pi f_c t + \phi_c)$$

$P_s = \text{watt}$

$$c(t) = \sqrt{2 \frac{E_b}{T_b}} \cos(2\pi f_c t + \phi_c)$$

where $E_b = \text{bit Energy}$
 $T_b = \text{bit duration}$

Practically

$$c(t) = \sqrt{2 \frac{E_b}{T_b}} \cos(2\pi f_c t + \phi_c)$$

where E_b = bit Energy
 T_b = bit duration

In practice ; assume unit energy measured over
one symbol (bit) duration
i.e $E_b = 1 \text{ J}$.

$$c(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t + \phi_c)$$

Using linear modulation scheme (BASK, BPSK, BFSK)

Modulated Signal

Modulated wave ; $S(t) = b(t) \cdot c(t)$

↓ Incoming binary wave

→ carrier wave (set $\phi_c = 0$ for convenience)

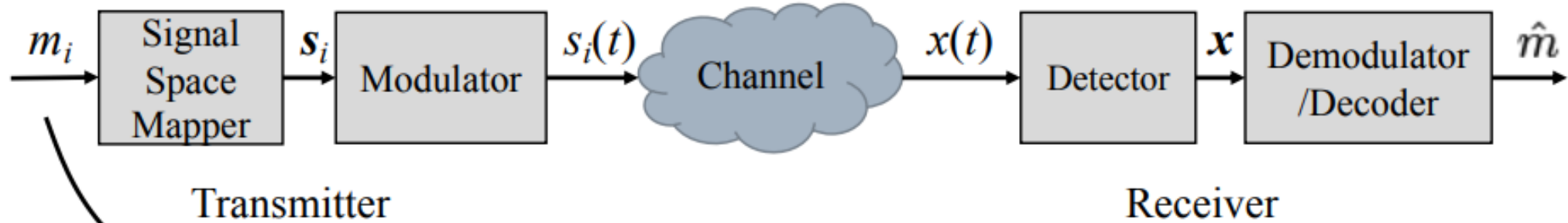
$$S(t) = \sqrt{\frac{2}{T_b}} b(t) \cos(2\pi f_c t)$$

Assuming

$f_c \gg W$ where

W = bandwidth of binary wave $b(t)$

(Real-valued) Passband Model – Message Source

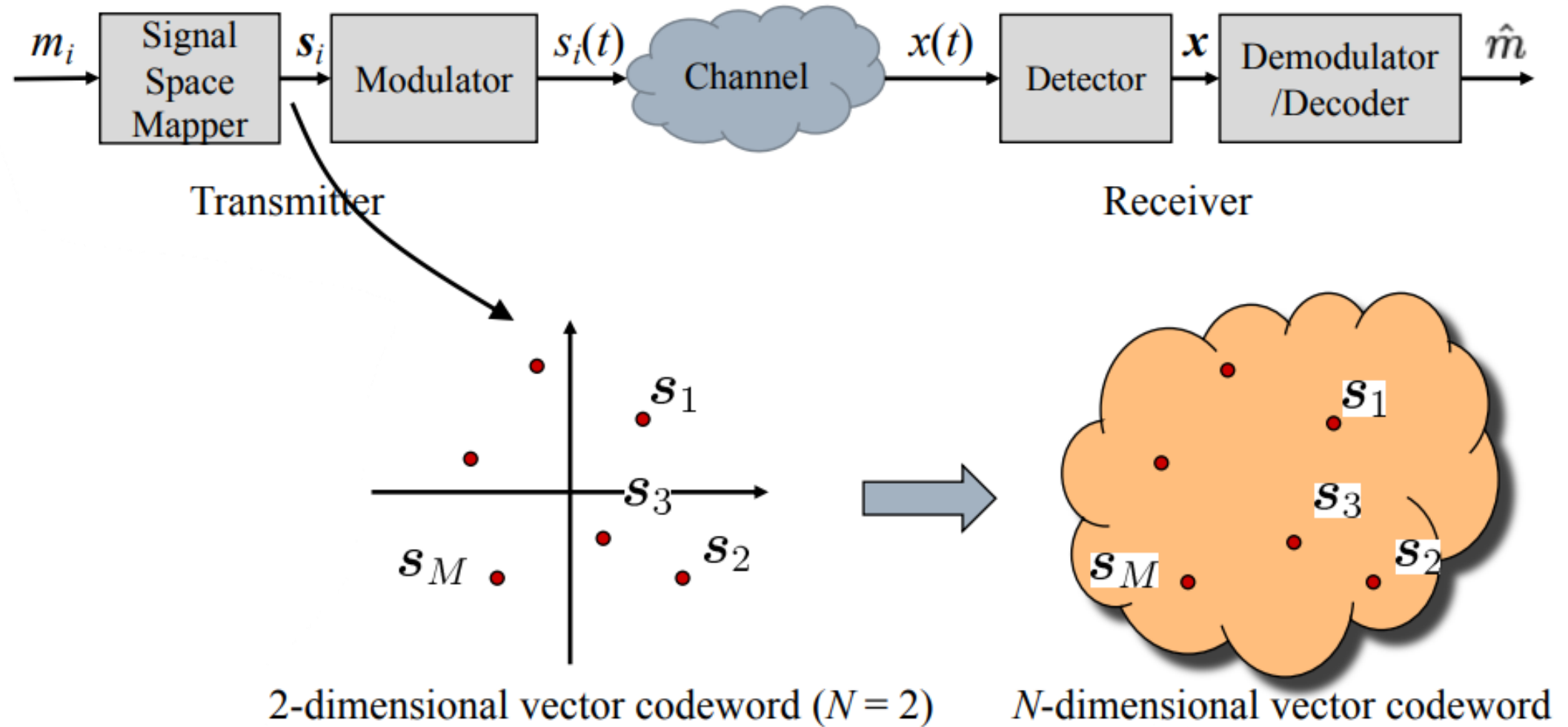


$$m_i \in \{m_1, m_2, \dots, m_M\}$$

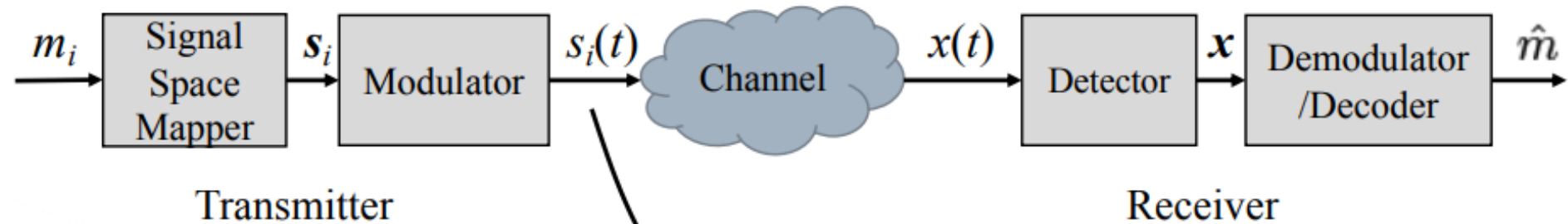
prior probability $p_i = P(m_i)$

$$\text{equal prior } p_i = \frac{1}{M}$$

(Real-valued) Passband Model – Signal Space Analysis



(Real-valued) Passband Model – Signal Space Analysis



$$\mathbf{s}_i = \begin{bmatrix} s_{i,1} \\ s_{i,2} \\ \vdots \\ s_{i,N} \end{bmatrix} \Rightarrow s_i(t) = \sum_{k=1}^N s_{i,k} \cdot \phi_k(t)$$

$\{\phi_k(t)\}_{k=1}^N$ is an orthonormal basis.

(Real-valued) Passband Model – Signal Space Analysis

□ $s_i(t)$ is an (finite) energy signal of duration T .

■ What is an energy signal?

Define the inner product of two signals $f(t)$ and $g(t)$ as

$$\langle f(t), g(t) \rangle = \int_0^T f(t)g(t)dt.$$

Then

$$\begin{aligned} \text{energy of signal } s_i(t) &= \langle s_i(t), s_i(t) \rangle \quad \left(= \|s_i(t)\|^2 \right) \\ &= \int_0^T s_i^2(t)dt < \infty \end{aligned}$$

(Real-valued) Passband Model – Signal Space Analysis

$$\begin{aligned}\langle s_i(t), s_i(t) \rangle &= \left\langle \sum_{k=1}^N s_{i,k} \cdot \phi_k(t), \sum_{\ell=1}^N s_{i,\ell} \cdot \phi_\ell(t) \right\rangle \\ &= \sum_{k=1}^N \sum_{\ell=1}^N \langle s_{i,k} \cdot \phi_k(t), s_{i,\ell} \cdot \phi_\ell(t) \rangle \\ &= \sum_{k=1}^N \sum_{\ell=1}^N s_{i,k} s_{i,\ell} \langle \phi_k(t), \phi_\ell(t) \rangle \\ &= \sum_{k=1}^N s_{i,k}^2\end{aligned}$$

(Real-valued) Passband Model – Signal Space Analysis

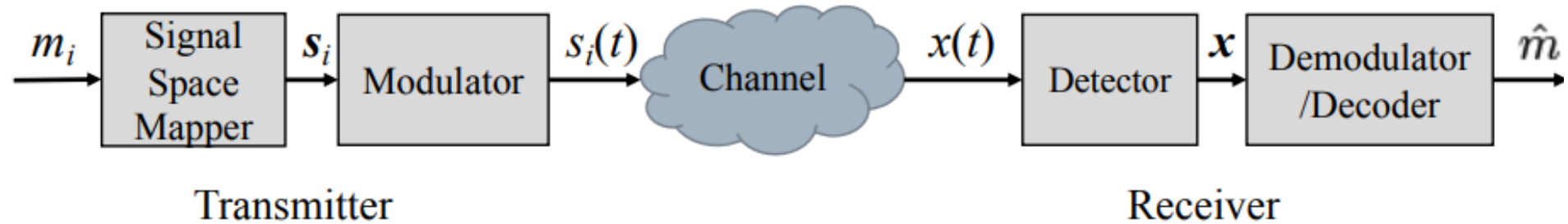


- Example of orthonormal basis $\{\phi_k(t)\}_{k=1}^N$ with $N = 2$ for ASK/PSK signals

$$\left\{ \sqrt{\frac{2}{T}} \cos(2\pi f_c t), \sqrt{\frac{2}{T}} \sin(2\pi f_c t) \right\}$$

with T being a multiple of $\frac{1}{f_c}$

(Real-valued) Passband Model – Signal Space Analysis

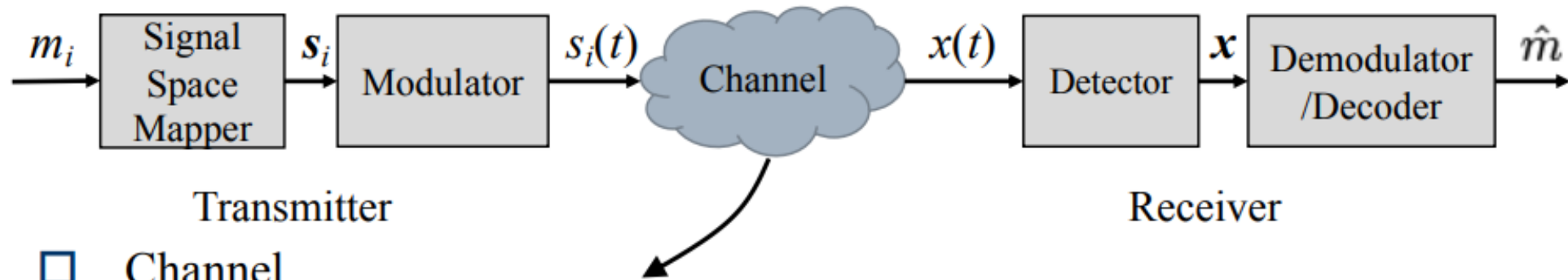


- Example of orthonormal baiss $\{\phi_k(t)\}_{k=1}^N$ with $N = 2$ for FSK signals

$$\left\{ \sqrt{\frac{2}{T}} \cos \left(2\pi \left(f_c - \frac{1}{2T} \right) t \right), \sqrt{\frac{2}{T}} \cos \left(2\pi \left(f_c + \frac{1}{2T} \right) t \right) \right\}$$

with T being a multiple of $\frac{1}{f_c}$

(Real-valued) Passband Model – Channel



□ Channel

- Linear: Principle of superposition

$$s_1(t) \mapsto x_1(t) \text{ and } s_2(t) \mapsto x_2(t) \Rightarrow as_1(t) + bs_2(t) \mapsto ax_1(t) + bx_2(t)$$

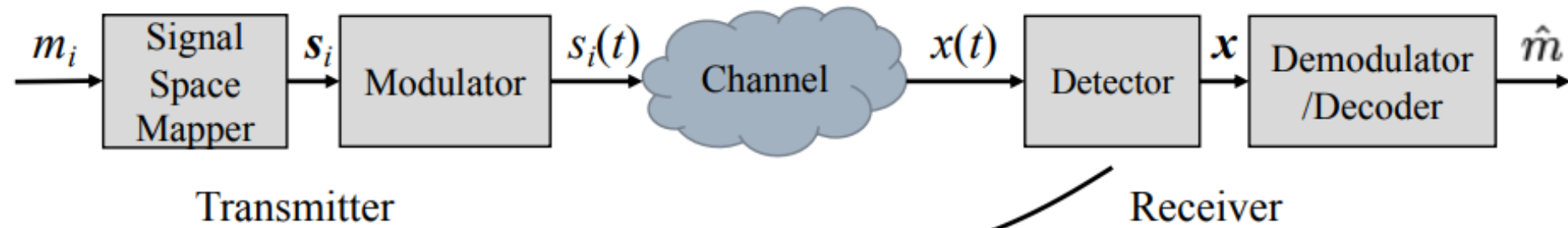
- Sufficient bandwidth

No loss of power in $s_i(t)$

- AWGN

$x(t) = s_i(t) + n(t)$, where $n(t)$ is a zero-mean white Gaussian process with two-sided power spectrum density $N_0/2$

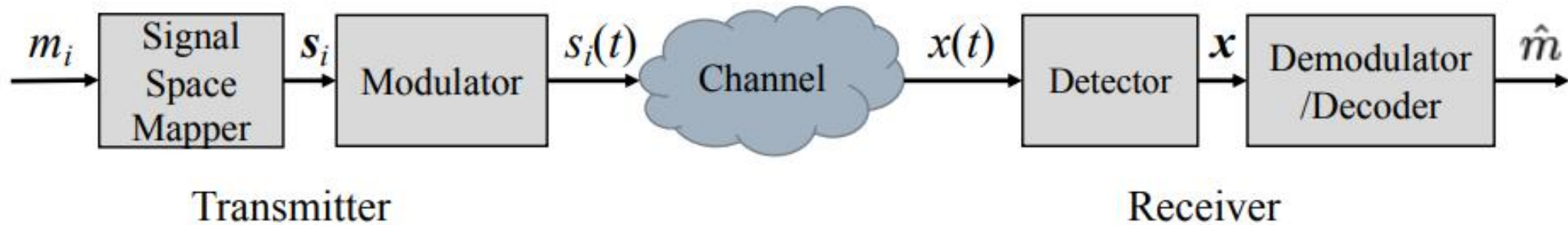
(Real-valued) Passband Model – Detector



□ Detector

$$x(t) = \sum_{k=1}^N x_k \cdot \phi_k(t) + \phi'(t) \quad \Rightarrow \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$
$$\left\langle \phi'(t), \sum_{k=1}^N x_k \cdot \phi_k(t) \right\rangle = 0$$

(Real-valued) Passband Model – Detector

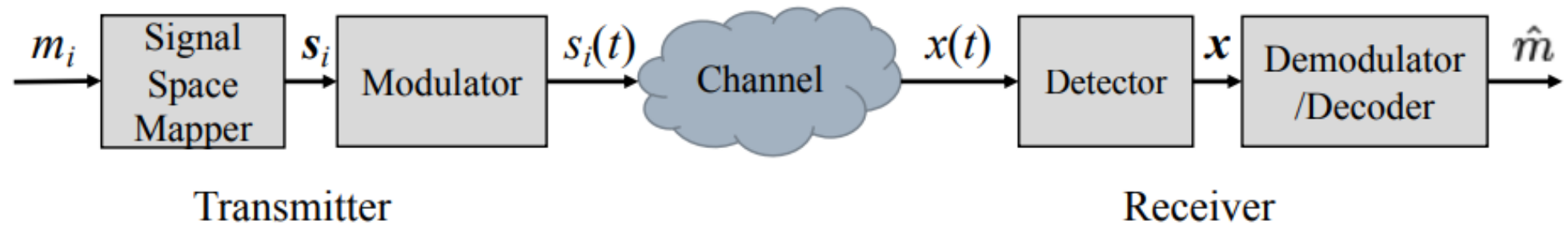


□ Demodulator/Decoder

$\hat{m}$ is the most probable transmitted message in $\{m_1, m_2, \dots, m_M\}$ given $\mathbf{x}$.

$$\hat{m} = \underbrace{\arg \max_{1 \leq i \leq M} P(m_i | \mathbf{x})}_{\text{maximum a posteriori (MAP)}} = \arg \max_{1 \leq i \leq M} \frac{P(m_i)}{P(\mathbf{x})} P(\mathbf{x} | m_i) = \underbrace{\arg \max_{1 \leq i \leq M} P(\mathbf{x} | m_i)}_{\text{maximum likelihood (ML)}}$$

Coherent Phase-Shift Keying (PSK) – Antipodal Signaling



□ Binary PSK

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)$$

$$s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \pi) = -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)$$

for $0 \leq t < T_b$, where T_b is a multiple of $1/f_c$.

Coherent Phase-Shift Keying (PSK) – Antipodal Signaling



□ Vector space analysis of binary PSK

■ Antipodal signal

$$\begin{aligned} s_1(t) &= +\sqrt{E_b} \cdot \phi_1(t) \\ s_2(t) &= -\sqrt{E_b} \cdot \phi_1(t) \end{aligned} \quad \Rightarrow \quad \begin{aligned} s_{11} &= \langle s_1(t), \phi_1(t) \rangle = +\sqrt{E_b} \\ s_{12} &= \langle s_2(t), \phi_1(t) \rangle = -\sqrt{E_b} \end{aligned}$$

$$\text{where } \phi_1(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t).$$

Coherent Phase-Shift Keying (PSK) – Antipodal Signaling

□ Error probability of binary PSK

$$x(t) = s(t) + w(t)$$

$$\Rightarrow \langle x(t), \phi_1(t) \rangle = \langle s(t), \phi_1(t) \rangle + \langle w(t), \phi_1(t) \rangle$$

$$\Rightarrow x = \pm\sqrt{E_b} + w$$

$$\Rightarrow \hat{m} = \arg \max \left\{ P(x | -\sqrt{E_b}), P(x | +\sqrt{E_b}) \right\}$$

$$\Rightarrow \hat{m} = \arg \max \left\{ \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x+\sqrt{E_b})^2/2\sigma^2}, \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\sqrt{E_b})^2/2\sigma^2} \right\}$$

$$\Rightarrow x \underset{+\sqrt{E_b}}{\overset{-\sqrt{E_b}}{\leq}} 0$$

$\sigma^2 = N_0/2$ is the variance of w

$$\Phi(-x) = \frac{1}{2} \operatorname{erfc} \left(\frac{x}{\sqrt{2}} \right)$$

Coherent Phase-Shift Keying (PSK) – Error probability

□ Error probability of binary PSK

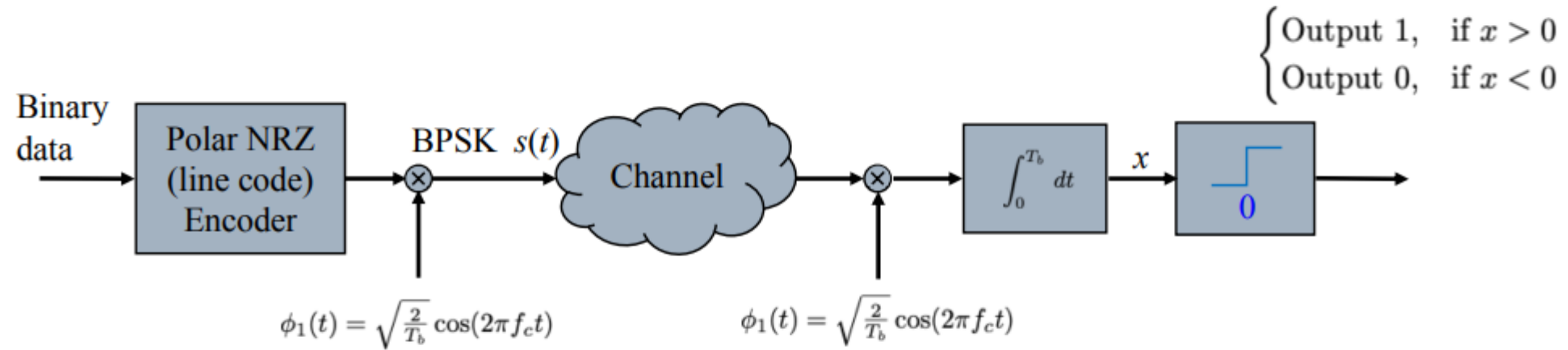
■ Based on the decision rule $x \underset{+\sqrt{E_b}}{\overset{-\sqrt{E_b}}{\leq}} 0$

$$\begin{aligned} P(\text{Error}) &= P\left(-\sqrt{E_b} \text{ transmitted}\right) P\left(x > 0 \mid -\sqrt{E_b} \text{ transmitted}\right) \\ &\quad + P\left(+\sqrt{E_b} \text{ transmitted}\right) P\left(x < 0 \mid +\sqrt{E_b} \text{ transmitted}\right) \\ &= \frac{1}{2} \int_0^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x+\sqrt{E_b})^2/2\sigma^2} dx + \frac{1}{2} \int_{-\infty}^0 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\sqrt{E_b})^2/2\sigma^2} dx \\ &= \int_{-\infty}^0 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\sqrt{E_b})^2/2\sigma^2} dx = \Phi\left(\frac{0 - \sqrt{E_b}}{\sigma}\right) = \Phi\left(-\sqrt{2\frac{E_b}{N_0}}\right) \end{aligned}$$

$$\Phi(-x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$$

Coherent Phase-Shift Keying (PSK) – Block diagram

□ Block diagram for PSK transmitter and (coherent) receiver



Thank You